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- Course Code & CO Reference: CSA0926, CO2
- Assignment Set: Direct-Descriptive
- Question:

Write a Java program that creates an ArrayList of integers and performs a sequence of operations on it: add several elements using add(), remove a specific element using remove(), retrieve an element at a given index using get(), and display the list's contents after each operation.

GitHub Repository Link: [https://github.com/geonaalis26-sudo/CSA0926-](https://github.com/geonaalis26-sudo/CSA0926-PROGRAMMING-IN-JAVA)

PROGRAMMING-IN-JAVA

- Submission Date: 01/08/2026

2. PROBLEM UNDERSTANDING

This program demonstrates basic operations of an ArrayList. It adds elements, removes one element, retrieves an element using its index, and displays the list after each operation.

3. APPROACH / DESIGN NOTES

- Imported java.util.ArrayList.
- Created an ArrayList<Integer>.
- Added elements using add().
- Removed an element using remove().
- Retrieved an element using get().
- Displayed the list after every operation.

4. SOURCE CODE

```
import java.util.ArrayList;
```

```
public class ArrayListOperations {
```

```

public static void main(String[] args) {

    ArrayList<Integer> list = new ArrayList<>();

    list.add(10);
    list.add(20);
    list.add(30);
    list.add(40);

    System.out.println("After adding elements: " + list);

    list.remove(1);

    System.out.println("After removing element: " + list);

    int value = list.get(1);

    System.out.println("Element at index 1: " + value);

    System.out.println("Final List: " + list);
}
}

```

5. TEST CASES

Test Case	Input	Expected Output	Actual Output
1 (Typical)	No input	List is created, element removed, element retrieved successfully	Same as expected
2 (Edge)	Remove last element	Last element removed correctly	Same as expected
3 (Boundary)	Get first element (index 0)	First element displayed correctly	Same as expected

6. SCREENSHOTS OF EXECUTION

Take screenshots of:

- Test Case 1: Output of the given program.

ArrayListOperations.java	Run	Output
<pre>1- import java.util.ArrayList; 2 3- public class ArrayListOperations { 4- public static void main(String[] args) { 5 6 ArrayList<Integer> list = new ArrayList<>(); 7 8 list.add(10); 9 list.add(20); 10 list.add(30); 11 list.add(40); 12 13 System.out.println("After adding elements: " + list); 14 15 list.remove(1); 16 17 System.out.println("After removing element: " + list); 18 19 int value = list.get(1); 20 21 System.out.println("Element at index 1: " + value); 22 23 System.out.println("Final List: " + list); 24 } 25 }</pre>		<pre>After adding elements: [10, 20, 30, 40] After removing element: [10, 30, 40] Element at index 1: 30 Final List: [10, 30, 40] === Code Execution Successful ===</pre>

- Test Case 2: Output after modifying the code to remove the last element.

ArrayListOperations.java	Run	Output
<pre>1- import java.util.ArrayList; 2 3- public class ArrayListOperations { 4- public static void main(String[] args) { 5 6 ArrayList<Integer> list = new ArrayList<>(); 7 8 list.add(10); 9 list.add(20); 10 list.add(30); 11 list.add(40); 12 13 System.out.println("After adding elements: " + list); 14 15 list.remove(3); 16 17 System.out.println("After removing element: " + list); 18 19 int value = list.get(1); 20 21 System.out.println("Element at index 1: " + value); 22 23 System.out.println("Final List: " + list); 24 } 25 }</pre>		<pre>After adding elements: [10, 20, 30, 40] After removing element: [10, 20, 30] Element at index 1: 20 Final List: [10, 20, 30] === Code Execution Successful ===</pre>

- Test Case 3: Output after modifying the code to retrieve the first element (get(0)).

ArrayListOperations.java	Run	Output
<pre>1- import java.util.ArrayList; 2 3- public class ArrayListOperations { 4- public static void main(String[] args) { 5 6 ArrayList<Integer> list = new ArrayList<>(); 7 8 list.add(10); 9 list.add(20); 10 list.add(30); 11 list.add(40); 12 13 System.out.println("After adding elements: " + list); 14 15 list.remove(3); 16 17 System.out.println("After removing element: " + list); 18 19 int value = list.get(0); 20 21 System.out.println("Element at index 1: " + value); 22 23 System.out.println("Final List: " + list); 24 } 25 }</pre>		<pre>After adding elements: [10, 20, 30, 40] After removing element: [10, 20, 30] Element at index 1: 10 Final List: [10, 20, 30] === Code Execution Successful ===</pre>

7. REFLECTION

I learned how to use the `ArrayList` class in Java. I understood the working of `add()`, `remove()`, and `get()` methods to manage list elements.

8. DECLARATION

This is my own work. AI assistance was used only to understand the problem, prepare the explanation, and format the solution.